Solution Requirements

Date	16 September 2026
Team ID	SWTID-2026-2630
Project Name	AI Fit Track API
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users register and log in with name, email & password; passwords are hashed with bcrypt.js and a JWT is issued on successful login.	High
2	Authorization levels	Every workout and AI route is protected by JWT-verification middleware (`middleware/auth.js`) that reads the Authorization header, verifies the token, and attaches the user to the request.	High
3	External interfaces	Integrates with Google Gemini AI via a dedicated `geminiService.js` for two flows: workout recommendation and fitness insights.	High
4	Transactions / data processing	Full CRUD plus search on **Workout** documents (name, category, duration, calories, date), each linked to its owning **User** via a reference.	High
5	Reporting	Returns structured JSON for every workout list/detail call and for every AI response, suitable for any client to render.	Medium
6	Business rules	Each user's email must be unique; workout category is restricted to a fixed enum (Cardio, Strength Training, Yoga, Running, Cycling, Walking); duration must be at least 1 minute; calories burned cannot be negative; a workout always belongs to exactly one authenticated user.	High
7	Compliance to laws or regulations	Passwords are never stored in plaintext (bcrypt hashing) or returned in API responses; JWT secret and the Gemini API key are kept in `.env` and excluded from version control via `.gitignore` .	High
8	Other	Centralized error handling (`middleware/errorHandler.js`) returns meaningful JSON error responses for invalid credentials, missing tokens, validation errors and database errors.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Auth and Workout endpoints (MongoDB-only) should respond quickly; AI Recommendation/Insights endpoints depend on Gemini's external latency.	Avg response < 200 ms (auth/workout CRUD); AI endpoints measured separately (see Performance Testing)
2	Scalability	Stateless JWT authentication allows multiple Node.js instances to run behind a load balancer with no shared session state.	Designed to support growth beyond a single-instance deployment
3	Security & Data Privacy	bcrypt password hashing, JWT-based sessionless auth, and Mongoose schema validation (required/min/enum) on every write.	No plaintext password storage; all inputs validated
4	Reliability & Availability	Centralized error-handling middleware plus a `config/db.js` connection guard that logs and exits on a failed MongoDB connection.	No unhandled crash on a bad request or a failed DB connection
5	Usability & Accessibility	Consistent, well-documented JSON REST responses, verified with a shared Thunder Client collection (`FitTrack.postman_collection.json`).	Every endpoint documented and testable from the shared collection
6	Other (Maintainability)	Modular structure (`config/ controllers/ middleware/ models/ routes/ services/`) under `src/` , so new endpoints can be added without touching existing modules.	New feature addable in a single new controller + route + (optional) service